Supplementary Material

Granular ball local consensus for graph-free node classification under label noise

Unless stated otherwise, accuracies in this supplement are percentages, reported as mean ± sample standard deviation over five paired seeds. Bold marks the highest mean within each dataset and noise setting. In Table C.1, each validation protocol is compared separately. Equations (H.1)–(H.6) are numbered within Appendix H; other equation references refer to the main paper.

Appendix A. Results of the main comparisons under all noise settings

A.1 Cora

Table A.1. Test accuracy on Cora under all five noise settings.

Method	Uniform 0.2	Uniform 0.4	Uniform 0.6	Pair 0.4	Instance 0.4
MLP+CE	54.96±2.22	44.24±2.62	32.51±3.38	42.83±2.19	45.33±2.22
Small-loss	59.84±1.29	51.95±1.97	36.72±1.09	46.99±2.77	51.82±0.94
Co-teaching	58.62±1.97	46.83±1.78	33.29±3.48	44.26±2.30	47.35±2.24
JoCoR	60.12±2.25	51.72±1.30	36.92±2.14	45.91±2.38	52.26±1.01
SCE	53.57±2.17	43.60±1.86	32.66±3.78	41.40±2.04	43.29±4.35
ELR	52.95±2.38	44.28±1.38	32.82±3.51	41.83±1.75	45.93±2.26
LogitClip	58.63±1.35	47.50±1.50	34.44±2.67	44.90±1.21	48.09±1.73
Random size-matched grouping	53.57±1.99	43.31±1.40	32.18±2.54	39.28±1.34	43.34±1.70
GB-LCSS	**63.25±2.46**	**57.49±0.76**	**41.86±2.30**	**51.45±2.80**	**56.84±1.67**

A.2 CiteSeer

Table A.2. Test accuracy on CiteSeer under all five noise settings.

Method	Uniform 0.2	Uniform 0.4	Uniform 0.6	Pair 0.4	Instance 0.4
MLP+CE	55.10±1.17	43.50±3.02	33.35±1.36	45.15±1.67	44.33±2.04
Small-loss	59.26±1.94	49.70±3.07	37.73±2.86	48.16±2.56	50.75±3.33
Co-teaching	57.05±0.82	46.24±2.89	34.19±2.92	46.25±2.29	46.81±1.68
JoCoR	59.36±1.43	49.91±3.37	39.07±2.47	48.63±2.35	50.86±2.45
SCE	52.12±3.28	42.09±3.47	30.59±2.83	42.22±1.56	40.92±4.23
ELR	53.62±1.48	43.17±3.95	30.80±2.70	42.69±2.43	41.94±3.27
LogitClip	58.75±1.44	46.57±3.52	34.47±1.69	45.78±1.06	47.41±2.64
Random size-matched grouping	55.13±1.85	42.08±2.88	30.36±1.88	43.92±1.67	43.07±2.00
GB-LCSS	**63.92±1.12**	**55.24±2.04**	**42.02±3.14**	**51.70±2.14**	**55.58±2.54**

A.3 CoraFull

Table A.3. Test accuracy on CoraFull under all five noise settings.

Method	Uniform 0.2	Uniform 0.4	Uniform 0.6	Pair 0.4	Instance 0.4
MLP+CE	41.11±0.46	31.98±0.98	21.16±0.81	28.14±1.17	31.33±0.85
Small-loss	41.58±0.94	33.04±0.50	21.51±0.98	30.15±1.11	32.90±0.87
Co-teaching	41.70±1.05	34.39±0.24	22.84±1.38	30.28±0.52	33.39±0.90
JoCoR	43.33±0.70	35.70±0.32	24.20±0.98	31.45±0.80	35.17±0.51
SCE	37.93±0.72	29.29±0.45	17.79±1.33	26.98±0.51	29.68±1.06
ELR	41.08±0.53	32.14±0.67	21.32±0.98	28.91±1.44	31.52±0.83
LogitClip	41.24±0.78	34.69±1.17	24.62±0.72	31.23±1.84	34.19±0.17
Random size-matched grouping	41.02±0.92	32.30±0.85	21.41±0.84	28.49±0.78	31.43±0.79
GB-LCSS	**43.46±0.57**	**37.39±0.51**	**27.43±0.89**	**34.46±1.23**	**36.95±0.53**

A.4 Coauthor-CS

Table A.4. Test accuracy on Coauthor-CS under all five noise settings.

Method	Uniform 0.2	Uniform 0.4	Uniform 0.6	Pair 0.4	Instance 0.4
MLP+CE	91.38±0.54	88.42±0.62	80.03±1.91	75.91±1.34	87.44±1.15
Small-loss	91.50±0.49	89.06±0.88	81.65±0.91	80.13±1.50	88.97±1.02
Co-teaching	91.70±0.29	88.80±0.97	81.88±0.72	79.06±1.30	88.95±0.97
JoCoR	**91.81±0.25**	89.38±0.61	82.58±0.41	81.02±1.62	89.17±0.85
SCE	90.63±0.47	88.37±0.53	79.43±1.45	79.77±2.51	87.99±0.56
ELR	91.21±0.60	88.29±0.73	80.54±1.28	78.03±1.48	87.78±0.94
LogitClip	91.32±0.43	88.40±0.21	79.96±1.64	81.57±0.53	88.60±0.75
Random size-matched grouping	90.26±0.70	87.46±1.07	78.78±1.82	75.99±0.60	86.87±1.56
GB-LCSS	91.54±0.26	**90.71±0.59**	**87.19±0.69**	**86.24±1.18**	**90.51±0.43**

A.5 Amazon-Photo

Table A.5. Test accuracy on Amazon-Photo under all five noise settings.

Method	Uniform 0.2	Uniform 0.4	Uniform 0.6	Pair 0.4	Instance 0.4
MLP+CE	82.34±1.39	73.90±1.80	58.60±3.77	62.61±4.31	69.13±3.00
Small-loss	84.96±0.58	77.25±1.62	61.40±3.72	66.70±5.73	73.00±1.25
Co-teaching	85.21±0.51	78.21±1.00	63.29±5.00	67.24±6.97	73.59±1.19
JoCoR	85.60±1.26	79.57±0.53	66.48±4.43	66.94±6.36	74.80±1.79
SCE	84.74±0.94	76.08±3.11	57.82±4.07	66.51±6.96	73.08±2.54
ELR	82.31±1.98	73.93±2.12	59.32±4.53	66.07±5.25	69.41±2.70
LogitClip	83.78±1.02	74.98±2.38	62.42±4.37	68.52±4.20	72.90±1.05
Random size-matched grouping	81.03±1.23	71.45±3.29	58.30±4.17	62.56±4.24	67.56±2.89
GB-LCSS	**86.07±1.05**	**82.12±1.08**	**71.45±3.75**	**73.89±3.73**	**79.59±0.88**

Appendix B. Paired statistical tests

For each setting, we compute the five seed-wise accuracy differences (GB-LCSS minus the reference method) and report their mean and 95% confidence interval in percentage points. Statistical significance is assessed with a two-sided paired t-test. Holm correction is applied separately to the 25 comparisons against each reference method. Confidence intervals are unadjusted, whereas the reported p-values are Holm-adjusted. Positive differences favor GB-LCSS; "Higher (seeds)" counts the seeds on which GB-LCSS has higher accuracy.

GB-LCSS has higher accuracy than JoCoR on all five seeds in 22 of the 25 settings.

Table B.1. Paired statistics for GB-LCSS against the three reference methods.

Dataset	Noise	Reference method	Mean difference (pp)	95% CI (pp)	Higher (seeds)	Adjusted p
Cora	Uniform 0.2	Random grouping	+9.68	[+7.16, +12.20]	5/5	0.006
Cora	Uniform 0.2	MLP+CE	+8.30	[+5.59, +11.00]	5/5	0.011
Cora	Uniform 0.2	JoCoR	+3.13	[+0.55, +5.72]	5/5	0.112
Cora	Uniform 0.4	Random grouping	+14.17	[+12.71, +15.63]	5/5	<0.001
Cora	Uniform 0.4	MLP+CE	+13.25	[+10.81, +15.68]	5/5	0.002
Cora	Uniform 0.4	JoCoR	+5.77	[+3.60, +7.95]	5/5	0.031
Cora	Uniform 0.6	Random grouping	+9.68	[+6.39, +12.96]	5/5	0.010
Cora	Uniform 0.6	MLP+CE	+9.35	[+4.54, +14.16]	5/5	0.011
Cora	Uniform 0.6	JoCoR	+4.94	[+1.79, +8.09]	5/5	0.082
Cora	Pair 0.4	Random grouping	+12.17	[+7.23, +17.11]	5/5	0.012
Cora	Pair 0.4	MLP+CE	+8.62	[+5.74, +11.50]	5/5	0.011
Cora	Pair 0.4	JoCoR	+5.54	[+2.90, +8.18]	5/5	0.060
Cora	Instance 0.4	Random grouping	+13.51	[+12.08, +14.94]	5/5	<0.001
Cora	Instance 0.4	MLP+CE	+11.51	[+9.50, +13.53]	5/5	0.002
Cora	Instance 0.4	JoCoR	+4.58	[+2.27, +6.89]	5/5	0.062
CiteSeer	Uniform 0.2	Random grouping	+8.79	[+6.87, +10.71]	5/5	0.004
CiteSeer	Uniform 0.2	MLP+CE	+8.81	[+7.00, +10.63]	5/5	0.003
CiteSeer	Uniform 0.2	JoCoR	+4.56	[+2.87, +6.26]	5/5	0.031
CiteSeer	Uniform 0.4	Random grouping	+13.16	[+11.26, +15.06]	5/5	<0.001
CiteSeer	Uniform 0.4	MLP+CE	+11.74	[+9.99, +13.50]	5/5	0.001
CiteSeer	Uniform 0.4	JoCoR	+5.34	[+2.51, +8.16]	5/5	0.062
CiteSeer	Uniform 0.6	Random grouping	+11.66	[+9.60, +13.72]	5/5	0.002
CiteSeer	Uniform 0.6	MLP+CE	+8.67	[+4.92, +12.42]	5/5	0.011
CiteSeer	Uniform 0.6	JoCoR	+2.95	[+1.09, +4.81]	5/5	0.082
CiteSeer	Pair 0.4	Random grouping	+7.78	[+5.04, +10.53]	5/5	0.010
CiteSeer	Pair 0.4	MLP+CE	+6.55	[+3.98, +9.13]	5/5	0.011
CiteSeer	Pair 0.4	JoCoR	+3.07	[+2.06, +4.07]	5/5	0.020
CiteSeer	Instance 0.4	Random grouping	+12.51	[+9.97, +15.05]	5/5	0.003
CiteSeer	Instance 0.4	MLP+CE	+11.25	[+8.48, +14.01]	5/5	0.006
CiteSeer	Instance 0.4	JoCoR	+4.72	[+3.63, +5.80]	5/5	0.007
CoraFull	Uniform 0.2	Random grouping	+2.44	[+1.48, +3.40]	5/5	0.012
CoraFull	Uniform 0.2	MLP+CE	+2.35	[+1.46, +3.25]	5/5	0.011
CoraFull	Uniform 0.2	JoCoR	+0.13	[-0.86, +1.12]	3/5	0.852
CoraFull	Uniform 0.4	Random grouping	+5.09	[+4.32, +5.87]	5/5	0.001
CoraFull	Uniform 0.4	MLP+CE	+5.41	[+4.51, +6.32]	5/5	0.002
CoraFull	Uniform 0.4	JoCoR	+1.69	[+0.86, +2.52]	5/5	0.062
CoraFull	Uniform 0.6	Random grouping	+6.02	[+5.16, +6.89]	5/5	<0.001
CoraFull	Uniform 0.6	MLP+CE	+6.27	[+5.52, +7.03]	5/5	<0.001
CoraFull	Uniform 0.6	JoCoR	+3.23	[+1.89, +4.57]	5/5	0.042
CoraFull	Pair 0.4	Random grouping	+5.98	[+4.75, +7.20]	5/5	0.003
CoraFull	Pair 0.4	MLP+CE	+6.32	[+4.66, +7.98]	5/5	0.007
CoraFull	Pair 0.4	JoCoR	+3.02	[+2.10, +3.93]	5/5	0.017
CoraFull	Instance 0.4	Random grouping	+5.52	[+4.23, +6.80]	5/5	0.004
CoraFull	Instance 0.4	MLP+CE	+5.62	[+4.37, +6.86]	5/5	0.004
CoraFull	Instance 0.4	JoCoR	+1.78	[+1.22, +2.34]	5/5	0.018
Coauthor-CS	Uniform 0.2	Random grouping	+1.28	[+0.70, +1.85]	5/5	0.012
Coauthor-CS	Uniform 0.2	MLP+CE	+0.16	[-0.33, +0.65]	3/5	0.418
Coauthor-CS	Uniform 0.2	JoCoR	-0.27	[-0.41, -0.13]	0/5	0.062

Dataset	Noise	Reference method	Mean difference (pp)	95% CI (pp)	Higher (seeds)	Adjusted p
Coauthor-CS	Uniform 0.4	Random grouping	+3.25	[+1.96, +4.55]	5/5	0.012
Coauthor-CS	Uniform 0.4	MLP+CE	+2.29	[+1.66, +2.93]	5/5	0.008
Coauthor-CS	Uniform 0.4	JoCoR	+1.33	[+1.07, +1.59]	5/5	0.004
Coauthor-CS	Uniform 0.6	Random grouping	+8.41	[+5.72, +11.10]	5/5	0.009
Coauthor-CS	Uniform 0.6	MLP+CE	+7.16	[+4.58, +9.75]	5/5	0.011
Coauthor-CS	Uniform 0.6	JoCoR	+4.61	[+3.37, +5.85]	5/5	0.011
Coauthor-CS	Pair 0.4	Random grouping	+10.25	[+8.67, +11.83]	5/5	0.001
Coauthor-CS	Pair 0.4	MLP+CE	+10.32	[+8.97, +11.67]	5/5	<0.001
Coauthor-CS	Pair 0.4	JoCoR	+5.22	[+3.82, +6.62]	5/5	0.011
Coauthor-CS	Instance 0.4	Random grouping	+3.65	[+2.12, +5.17]	5/5	0.012
Coauthor-CS	Instance 0.4	MLP+CE	+3.07	[+2.15, +4.00]	5/5	0.009
Coauthor-CS	Instance 0.4	JoCoR	+1.34	[+0.76, +1.92]	5/5	0.045
Amazon-Photo	Uniform 0.2	Random grouping	+5.03	[+3.82, +6.25]	5/5	0.005
Amazon-Photo	Uniform 0.2	MLP+CE	+3.72	[+2.63, +4.81]	5/5	0.009
Amazon-Photo	Uniform 0.2	JoCoR	+0.46	[-0.99, +1.92]	3/5	0.852
Amazon-Photo	Uniform 0.4	Random grouping	+10.66	[+6.55, +14.78]	5/5	0.012
Amazon-Photo	Uniform 0.4	MLP+CE	+8.22	[+5.17, +11.26]	5/5	0.011
Amazon-Photo	Uniform 0.4	JoCoR	+2.55	[+1.02, +4.07]	5/5	0.078
Amazon-Photo	Uniform 0.6	Random grouping	+13.15	[+9.63, +16.67]	5/5	0.006
Amazon-Photo	Uniform 0.6	MLP+CE	+12.85	[+8.39, +17.31]	5/5	0.011
Amazon-Photo	Uniform 0.6	JoCoR	+4.96	[+0.60, +9.32]	5/5	0.112
Amazon-Photo	Pair 0.4	Random grouping	+11.32	[+8.02, +14.63]	5/5	0.008
Amazon-Photo	Pair 0.4	MLP+CE	+11.28	[+9.10, +13.46]	5/5	0.003
Amazon-Photo	Pair 0.4	JoCoR	+6.95	[+1.90, +12.00]	5/5	0.094
Amazon-Photo	Instance 0.4	Random grouping	+12.03	[+8.45, +15.61]	5/5	0.008
Amazon-Photo	Instance 0.4	MLP+CE	+10.46	[+6.95, +13.98]	5/5	0.011
Amazon-Photo	Instance 0.4	JoCoR	+4.79	[+2.43, +7.15]	5/5	0.062

Appendix C. Sensitivity to noisy validation labels

This experiment checks whether noisy validation labels affect model selection under uniform 0.4 noise. The training split and training-label noise are kept unchanged. Validation labels are independently corrupted with the same uniform noise rate using a separate random seed, while test labels remain unchanged. GB-LCSS and random size-matched grouping use the target construction in Section 3.3; GSDN and PosteL follow the implementations described in Section 4.1.3 and Appendix F.

Table C.1. Test accuracy under clean and noisy validation labels.

Dataset	Method	Clean validation	Noisy validation
Cora	GB-LCSS	**57.49±0.76**	**56.81±1.55**
Cora	Random size-matched grouping	43.31±1.40	43.12±2.07
Cora	GSDN	55.97±4.29	54.81±4.70
Cora	MLP+CE	44.24±2.62	43.30±3.65
Cora	JoCoR	51.72±1.30	50.77±1.98
Cora	PosteL	47.09±1.81	46.81±1.94

Dataset	Method	Clean validation	Noisy validation
CiteSeer	GB-LCSS	**55.24±2.04**	**54.99±2.18**
CiteSeer	Random size-matched grouping	42.08±2.88	42.65±2.49
CiteSeer	GSDN	50.36±3.39	50.60±1.84
CiteSeer	MLP+CE	43.50±3.02	43.06±2.96
CiteSeer	JoCoR	49.91±3.37	50.67±1.07
CiteSeer	PosteL	45.84±2.48	46.27±1.76
CoraFull	GB-LCSS	**37.39±0.51**	**36.90±0.81**
CoraFull	Random size-matched grouping	32.30±0.85	32.03±0.77
CoraFull	GSDN	35.80±0.66	36.13±0.69
CoraFull	MLP+CE	31.98±0.98	31.99±0.85
CoraFull	JoCoR	35.70±0.32	35.57±1.04
CoraFull	PosteL	36.51±1.01	35.74±1.14

Under noisy validation labels, GB-LCSS leads random size-matched grouping by 13.69, 12.35, and 4.87 percentage points on Cora, CiteSeer, and CoraFull, respectively, and leads MLP+CE by 13.51, 11.93, and 4.91 points. The ranking of the six methods is unchanged on Cora (Kendall's tau = 1.000) and remains similar on CiteSeer and CoraFull (tau = 0.867).

Appendix D. Additional target-construction analyses

Table 5 gives the uniform 0.4 vote and smoothing controls. Averaged over the five noise settings, the accuracy gain from excluding the current node’s vote ranges from 0.08 to 1.03 percentage points across datasets, with a minimum setting-wise difference of −0.49 points.

With clean training labels and otherwise unchanged settings, the accuracy differences from MLP+CE are −0.24, +2.00, −0.38, −0.87, and −1.75 percentage points on Cora, CiteSeer, CoraFull, Coauthor-CS, and Amazon-Photo, respectively.

Ties among the most-voted classes occur in 123 of the 125 dataset-noise-seed runs and in 911 ball instances, affecting 12,788 of 259,000 training-node targets (4.94%). When several classes tie for the most votes, we keep the full vote distribution instead of choosing one tied class. This prevents the result from depending on arbitrary class numbering.

Appendix E. Results of graph-based soft-target methods under all noise settings

Table E.1 reports the five-seed means and sample standard deviations of GB-LCSS, PosteL, ALS, and SALS on the five main datasets and five noise settings. All four methods use the same data splits, training-label noise realizations, and graph-free inference protocol. Implementation details are given in Appendix F.

Table E.1. Test accuracy of graph-based soft-target methods under all five noise settings.

Dataset	Noise setting	GB-LCSS	PosteL	ALS	SALS
Cora	Uniform 0.2	**63.25±2.46**	58.36±1.64	55.98±1.42	58.23±1.67
Cora	Uniform 0.4	**57.49±0.76**	47.09±1.81	45.20±1.24	48.27±2.50
Cora	Uniform 0.6	**41.86±2.30**	32.33±2.37	32.72±2.54	35.34±1.74
Cora	Pair 0.4	**51.45±2.80**	44.36±0.65	41.63±3.61	44.16±1.52
Cora	Instance 0.4	**56.84±1.67**	46.62±2.17	45.73±1.34	48.54±1.17

Dataset	Noise setting	GB-LCSS	PosteL	ALS	SALS
CiteSeer	Uniform 0.2	**63.92±1.12**	58.11±1.11	56.58±2.04	58.36±1.07
CiteSeer	Uniform 0.4	**55.24±2.04**	45.84±2.48	43.77±2.76	47.20±1.69
CiteSeer	Uniform 0.6	**42.02±3.14**	31.88±0.86	32.00±1.57	35.16±2.58
CiteSeer	Pair 0.4	**51.70±2.14**	45.06±1.60	45.72±2.41	45.63±2.70
CiteSeer	Instance 0.4	**55.58±2.54**	46.02±2.54	43.75±2.18	47.51±2.50
CoraFull	Uniform 0.2	43.46±0.57	**44.54±0.31**	41.72±0.22	44.43±0.75
CoraFull	Uniform 0.4	**37.39±0.51**	36.51±1.01	31.91±1.05	37.06±0.78
CoraFull	Uniform 0.6	**27.43±0.89**	23.90±1.16	20.64±0.61	26.10±1.07
CoraFull	Pair 0.4	**34.46±1.23**	32.62±1.71	29.83±0.92	32.85±1.68
CoraFull	Instance 0.4	**36.95±0.53**	35.99±0.51	31.15±1.18	36.58±0.74
Coauthor-CS	Uniform 0.2	**91.54±0.26**	89.01±0.40	91.13±0.57	91.08±0.49
Coauthor-CS	Uniform 0.4	**90.71±0.59**	84.74±1.51	88.55±0.86	88.85±0.82
Coauthor-CS	Uniform 0.6	**87.19±0.69**	74.61±1.95	79.36±1.67	82.99±0.95
Coauthor-CS	Pair 0.4	**86.24±1.18**	78.45±2.28	74.99±2.24	77.85±1.51
Coauthor-CS	Instance 0.4	**90.51±0.43**	85.72±1.35	87.53±1.27	88.33±0.66
Amazon-Photo	Uniform 0.2	**86.07±1.05**	84.37±0.96	82.63±1.27	84.81±1.20
Amazon-Photo	Uniform 0.4	**82.12±1.08**	80.59±0.94	73.41±2.75	77.86±0.79
Amazon-Photo	Uniform 0.6	**71.45±3.75**	64.49±2.99	58.14±1.78	66.29±3.62
Amazon-Photo	Pair 0.4	**73.89±3.73**	67.56±3.57	62.82±6.11	64.83±3.54
Amazon-Photo	Instance 0.4	**79.59±0.88**	77.23±2.77	68.81±2.81	72.73±1.78

Appendix F. Implementation details and key settings

Table F.1 summarizes how the external graph-based and distillation methods are implemented in our experiments. All methods use the same data splits and noise realizations, and the deployed MLP receives only node attributes. Table F.2 lists the method-specific settings used in the main comparisons.

Table F.1. Implementation used for the external graph-based and distillation methods.

Method	Implementation used	Main component used
PosteL	Official target code integrated	Posterior soft-target construction; iterative pseudo-labeling disabled
ALS	Adapted implementation	Adaptive label smoothing, pacing, and KL term
SALS	Adapted implementation	Structure-aware label smoothing
Graph-MLP	Adapted to the common MLP	Multi-hop NContrast objective
GSDN	Adapted implementation	Feature- and label-level neighborhood self-distillation
GCN(X)+CE teacher	Project implementation	GCN-to-MLP response distillation
FF-G2M	Reimplemented with common backbones	Low- and high-frequency distillation
KRD	Adapted implementation	Reliability-based knowledge selection
VQGraph	Adapted implementation	Soft-codebook distillation

Other optimizer and MLP settings follow Section 4.1.4.

Table F.2. Key method-specific settings used in the main comparisons.

Method	Key settings	Source
Small-loss	Forget rate = known synthetic noise rate; 10-epoch linear ramp-up; exponent 1.0	Project setting
Co-teaching	Two independent MLPs; same forget-rate schedule as Small-loss	NoisyGL-based
JoCoR	Two independent MLPs; same forget-rate schedule as Small-loss; consistency weight 0.1	NoisyGL-based
SCE	Cross-entropy weight 0.1; reverse-cross-entropy weight 1.0	NoisyGL-based

Method	Key settings	Source
ELR	Moving-average coefficient 0.7; regularization weight 1.0	Original objective
LogitClip	Logit-clipping threshold 1.0	Original objective
PosteL	Soft-label ratio 0.8; smoothing ratio 0.4; degree cutoff 1; temperature 1.0; iterative pseudo-labeling disabled	Official target code
ALS	2 propagation steps; residual coefficient 0.5; KL weight 1e-3; exponential pacing; maximum smoothing weight 0.1	Fixed project setting
SALS	epsilon = 0.4; gamma = 0.8; unlabeled neighborhoods use the uniform distribution	Fixed project setting
Graph-MLP	Graph batch size 512; neighborhood-contrastive weight 0.5; temperature 0.5	Fixed project setting
GSDN	Graph batch size 512; feature- and label-level weights 0.8; negative-constraint weight 0.1; initial mixup coefficient 0.5	Fixed project setting
GB-LCSS	Bmax = 64, nmin = 32, delta = 0.05, smin = 2, beta = 0.5; current node excluded from voting; ties keep the empirical distribution; 2-means uses Euclidean distance, k-means++, and 10 restarts	Set during method development on Cora and CiteSeer; used unchanged in the main comparisons
GCN(X)+CE teacher	Two-layer GCN teacher and two-layer MLP student, both width 64; 200 epochs; temperature 1.0	GLNN-based
FF-G2M	Two-layer GCN teacher, width 64; low- and high-frequency distillation weights 0.4; temperature 1.5	Public objective
KRD	4 feature perturbations, standard deviation 0.01; top 50% of nodes retained by reliability; temperature 1.0	Public mechanism
VQGraph	128 soft codes; codebook temperature 1.0; token/node/edge weights 1.0/0.001/0.03; 4096 edges sampled per epoch	Public mechanism
GCN teacher trained on GB-LCSS targets	Two-layer GCN(X) teacher, width 64; GB-LCSS targets from Eq. (14); student temperature 1.0; student fits teacher predictions over all nodes	Proposed extension

The settings above were fixed before the full comparison grid and then kept unchanged across datasets. Complete configurations are provided in the supplementary repository.

Table F.3 lists the GRACE settings used for the five main datasets. Cora and CiteSeer use their public dataset settings; CoraFull and Coauthor-CS use the DBLP setting, and Amazon-Photo uses the PubMed setting. The Dimensions column gives the hidden/projection dimensions. Paired edge-dropping and feature-masking values correspond to the two augmented views. All configurations use a two-layer GCN encoder with weight decay 1e-5.

Table F.3. GRACE settings for the five main datasets.

Dataset	Configuration	Dimensions	Learning rate	Epochs	Edge dropping	Feature masking	Temperature
Cora	Public Cora config	128/128	1e-4	200	0.2/0.4	0.3/0.4	0.4
CiteSeer	Public CiteSeer config	256/256	1e-3	200	0.2/0.0	0.3/0.2	0.9
CoraFull	Public DBLP config	256/256	1e-3	1000	0.1/0.4	0.1/0.0	0.7
Coauthor-CS	Public DBLP config	256/256	1e-3	1000	0.1/0.4	0.1/0.0	0.7
Amazon-Photo	Public PubMed config	256/256	1e-3	1500	0.4/0.1	0.0/0.2	0.7

These settings were fixed before the label-noise experiments. GRACE pretraining does not use labels or data-split masks.

Appendix G. Results on additional datasets

The six additional datasets are evaluated under uniform 0.4 noise. We also evaluate CiteSeerFull under the other four noise settings in an exploratory follow-up. Table G.1 summarizes their basic statistics; data processing and edge homophily follow Table 2.

Table G.1. Basic statistics of the six additional datasets.

Dataset	Nodes	Undirected edges	Features	Classes	Training nodes	Edge homophily
PubMed	19,717	44,324	500	3	3,943	0.802
Amazon-Computers	13,752	245,861	767	10	2,750	0.777
Cora_ML	2,995	8,158	2,879	7	599	0.789
Attributed-Wiki	2,405	11,596	4,973	17	481	0.610
CiteSeerFull	4,230	5,337	602	6	846	0.949
Coauthor-Physics	34,493	247,962	8,415	5	6,898	0.931

Table G.2 compares GB-LCSS with the strongest of the seven noisy-label baselines in Table 3 and with random size-matched grouping on the six additional datasets under uniform 0.4 noise. Differences are GB-LCSS minus the corresponding method, in percentage points; "Higher (seeds)" counts the seeds on which GB-LCSS has higher accuracy than the strongest baseline. The same definitions apply to Table G.3.

Table G.2. Uniform 0.4 results on the six additional datasets.

Dataset	GB-LCSS	Best baseline	Baseline accuracy	Difference vs. baseline	Higher (seeds)	Random grouping	Difference vs. random
PubMed	72.60±0.66	JoCoR	**77.84±1.10**	-5.24	0/5	63.04±4.49	+9.56
Amazon-Computers	76.20±0.56	JoCoR	**76.81±0.85**	-0.60	1/5	70.49±1.08	+5.71
Cora_ML	**56.26±2.61**	Small-loss	55.56±3.42	+0.70	3/5	47.81±3.34	+8.45
Attributed-Wiki	65.17±1.42	LogitClip	**67.83±1.51**	-2.66	0/5	59.83±2.13	+5.34
CiteSeerFull	**60.94±1.08**	JoCoR	59.48±1.60	+1.46	4/5	46.69±2.74	+14.25
Coauthor-Physics	**93.13±0.25**	JoCoR	93.09±0.28	+0.04	3/5	88.80±2.12	+4.33

GB-LCSS has higher mean accuracy than random grouping on all six datasets by 4.33–14.25 percentage points and has higher mean accuracy than the strongest noisy-label baseline on three. The difference from the strongest baseline ranges from -5.24 points on PubMed to +1.46 points on CiteSeerFull. Table G.3 reports all five noise settings for CiteSeerFull; GB-LCSS has mean accuracy at least as high as the strongest baseline in four settings, with a 4.20-point deficit under uniform 0.6.

Table G.3. CiteSeerFull results under all five noise settings.

Noise setting	GB-LCSS	Best baseline	Baseline accuracy	Difference vs. baseline	Higher (seeds)	Random grouping
Uniform 0.2	**70.71±0.65**	Small-loss	67.47±1.98	+3.24	5/5	58.70±1.88
Uniform 0.4	**60.94±1.08**	JoCoR	59.48±1.60	+1.46	4/5	46.69±2.74
Uniform 0.6	42.45±3.08	JoCoR	**46.65±3.70**	-4.20	0/5	33.13±3.53
Pair 0.4	**55.02±4.07**	JoCoR	51.12±5.88	+3.89	4/5	43.86±3.51
Instance 0.4	**61.49±3.01**	JoCoR	59.92±4.19	+1.57	3/5	47.51±2.21

Appendix H. Analysis of local-consensus supervision

This appendix proves Proposition 1 and derives the soft-target error identity in Section 3.6. All probabilities in Section H.1 condition on the graph, frozen representation, granular ball assignments, training-node set, and clean labels. Randomness comes from the label-corruption process, not from resampling graph nodes.

H.1 Reliability of the local vote

Fix a training node v in ball B_j, with $s_v = |T_{j,-v}| \geq s_{\min}$. The clean class proportions $\pi_{v,c}$ and the target-class margin Δ_v are defined in Eq. (21). Each voting label independently stays unchanged with probability $1-\eta$ or changes to any particular incorrect class with probability $\eta/(C-1)$. For $\gamma = 1 - C\eta/(C-1)$, the expected empirical vote proportion is

$$\mathbb{E}[p_{j,-v,c}] = (1-\eta)\pi_{v,c} + \frac{\eta}{C-1}(1-\pi_{v,c}) = \gamma\pi_{v,c} + \frac{\eta}{C-1}. \tag{H.1}$$

Therefore, the expected vote difference between the current node's true class and any competing class $c \neq y_v$ is

$$\mathbb{E}[p_{j,-v,y_v} - p_{j,-v,c}] = \gamma(\pi_{v,y_v} - \pi_{v,c}) \geq \gamma\Delta_v. \tag{H.2}$$

When $\gamma > 0$, symmetric noise preserves the ordering of the clean class proportions in expectation. This does not require the true class to hold more than half of the votes: it requires a positive margin over every individual competing class.

For each competing class, define $Z_{u,c} = \mathbb{I}(\tilde{y}_u = y_v) - \mathbb{I}(\tilde{y}_u = c)$ for $u \in T_{j,-v}$. The variables $Z_{u,c}$ are independent across voting nodes, lie in $[-1,1]$, and need not be identically distributed. Their average equals $p_{j,-v,y_v} - p_{j,-v,c}$. Applying Hoeffding's inequality [H1] gives

$$\begin{aligned} \Pr(p_{j,-v,c} \geq p_{j,-v,y_v}) &\leq \exp\left[-\frac{s_v\gamma^2(\pi_{v,y_v}-\pi_{v,c})^2}{2}\right] \\ &\leq \exp\left(-\frac{s_v\gamma^2\Delta_v^2}{2}\right). \end{aligned} \tag{H.3}$$

A union bound over the $C-1$ competing classes, capped at one, proves Eq. (22). A tie is included in the failure event. If the empirical winner is unique, Eq. (10) sharpens that winner without changing its identity; if several classes tie, it retains the empirical distribution. Thus the maximizing class set is unchanged for every $\beta \in [0,1]$, and the same probability bound applies to $q_{j,-v}$.

The conditioning keeps the clean class composition fixed; it does not assume that clean graph labels are independent. Independent label corruption is the required probabilistic assumption. The positive-margin condition is node-specific: a ball's dominant clean class need not be the class of every member. With a nonpositive margin, increasing the number of votes can reinforce evidence for a competing class. With few votes or a small positive margin, the bound may equal one and provide no informative numerical guarantee. The symmetric-noise ordering argument does not extend directly to pair or instance-dependent noise.

H.2 Error of the consensus-weighted target

Fix any realization of the labels. For a node with sufficient voting support, use the abbreviations $z = e_{\tilde{y}_v}$, $q = q_{j,-v}$, $\alpha = \alpha_{j,-v}$, and $e = e_{y_v}$. Expanding the square gives

$$\begin{aligned} \| t_v - e\|_2^2 = \; & (1-\alpha)^2 \| z - e\|_2^2 + \alpha^2 \| q - e\|_2^2 \\ & + 2\alpha(1-\alpha)\langle z - e, q - e\rangle. \end{aligned} \tag{H.4}$$

Substituting $2\langle z-e, q-e\rangle = \| z-e\|_2^2 + \| q-e\|_2^2 - \| z-q\|_2^2$ yields Eq. (23). This identity is pointwise, so it remains valid when α and q depend on the same local votes. For a node below the support threshold, Eq. (12) retains z, giving zero change in target error.

If the observed label is correct, $z = e$ and $L_v(t_v) = \alpha^2 L_v(q)$. If it is corrupted, $L_v(z) = 2$, and the soft target can lower this error. Consequently, the change in the mean Brier score can be written exactly as

$$\begin{aligned} \mathrm{BS}(Z) - \mathrm{BS}(T) = \quad & \frac{1}{|\mathcal{V}_L|} \sum_{v \in \mathcal{N}} (2 - L_v(t_v)) \\ & - \frac{1}{|\mathcal{V}_L|} \sum_{v \in \mathcal{K}} L_v(t_v), \end{aligned} \tag{H.5}$$

where $\mathcal{N} = \{v \in \mathcal{V}_L : \tilde{y}_v \neq y_v\}$ is the set of corrupted training labels and $\mathcal{K} = \mathcal{V}_L \setminus \mathcal{N}$. The matrices Z and T contain the observed one-hot targets and the GB-LCSS targets, respectively; the Brier convention is the sum over classes, averaged over training nodes. Equation (H.5) separates improvement on corrupted labels from the perturbation of correct labels. A reduction in squared probability error need not change the most-probable class of a target.

A conditional sufficient condition follows under the independent symmetric-noise model. Hold the other observed training labels fixed, as well as the quantities conditioned on in Section H.1. Excluding the current node means that q and α do not change when only its observed label is varied. Taking expectation over this node's label corruption gives $\mathbb{E}_v L_v(z) = 2\eta$. Equation (23) then implies

$$\mathbb{E}_v L_v(t_v) \leq (1 - \alpha)\, 2\eta + \alpha L_v(q). \tag{H.6}$$

Hence $L_v(q) < 2\eta$ and $\alpha > 0$ are sufficient for lower conditional expected target error than the observed one-hot label. The condition concerns the quality of the local distribution, not merely its concentration. Proposition 1 controls which class receives the most local votes; it does not by itself establish this squared-error condition. The Brier comparisons in Section 4.3.1 evaluate target quality, while the classification experiments assess the performance of the trained MLP.

Reference for Appendix H

[H1] W. Hoeffding, Probability inequalities for sums of bounded random variables, J. Am. Stat. Assoc. 58 (1963) 13–30.

Appendix I. Parameter sensitivity and target-error decomposition

I.1 Parameter sensitivity and representation controls

These experiments use Cora, CiteSeer, and CoraFull with uniform 0.4 noise on training labels, a 20%/10%/70% split, clean validation labels, and seeds 0–4. Each dataset retains the frozen GRACE representation used in the main comparisons. Each parameter variant is paired with the default configuration using the same split, noise realization, and initialization seed. The target rule, MLP architecture, optimizer, 200-epoch budget, and validation-based checkpoint selection remain unchanged.

Targets were computed on the CPU, and the MLP was trained on an AMD Radeon RX 7700 XT using PyTorch 2.9.1+rocm7.2.1. The scikit-learn 1.8.0 implementation of 2-means used six CPU threads; PyTorch used four host threads. The default configuration was rerun with this setup and serves as the reference for all variants in Table 7 and Table I.1.

The default values are $\beta = 0.5$, $B_{\max} = 64$, $n_{\min} = 32$, $\delta = 0.05$, and $s_{\min} = 2$. Each sensitivity row changes only the indicated parameter. The voting threshold $s_{\min}$ applies after the current node is excluded. Setting $\beta = 0$ removes the extra weight on the most-voted class but retains the empirical local distribution. Random grouping preserves the default ball sizes; raw X and A^2X replace the representation used for grouping, as described in Section 4.3.3.

Table I.1. Parameter sensitivity and controls under uniform 0.4 noise. Accuracy is mean ± sample standard deviation over five paired seeds.

Configuration	Cora	CiteSeer	CoraFull
Default	57.52 ± 0.71	55.24 ± 2.04	37.22 ± 0.66
$\beta = 0$	55.13 ± 1.88	52.94 ± 3.67	37.11 ± 0.36
$\beta = 0.25$	56.44 ± 1.49	54.47 ± 2.63	37.36 ± 0.44
$\beta = 0.75$	57.59 ± 1.16	56.36 ± 1.83	37.02 ± 1.34
$\beta = 1$	57.56 ± 1.11	56.98 ± 2.07	37.08 ± 1.18
$B_{\max} = 32$	57.55 ± 1.09	56.26 ± 1.93	35.12 ± 0.61
$B_{\max} = 128$	57.52 ± 0.71	55.46 ± 2.07	38.79 ± 0.14
$s_{\min} = 5$	56.09 ± 1.43	54.74 ± 2.47	37.24 ± 0.67
$s_{\min} = 10$	50.25 ± 1.48	50.02 ± 1.92	37.12 ± 0.65
$n_{\min} = 1$	55.87 ± 1.71	55.94 ± 1.80	37.17 ± 0.66
$n_{\min} = 16$	56.58 ± 1.53	56.49 ± 1.18	36.92 ± 0.59
$n_{\min} = 64$	57.45 ± 1.30	56.23 ± 2.78	37.60 ± 0.57
$\delta = 0.01$	57.52 ± 0.71	55.24 ± 2.04	37.22 ± 0.66
$\delta = 0.10$	44.37 ± 2.31	42.90 ± 2.50	31.86 ± 1.05
MLP+CE	44.24 ± 2.62	43.50 ± 3.02	31.98 ± 0.98
Random size-matched grouping	43.31 ± 1.40	42.00 ± 2.91	32.30 ± 0.85
Raw X	44.37 ± 2.31	42.90 ± 2.50	31.86 ± 1.05
A^2X	44.37 ± 2.31	42.90 ± 2.50	31.86 ± 1.05

Configurations with identical target tensors for the same dataset and seed shared one training result. The 270 evaluated combinations therefore comprise 214 training runs and 56 reuses after exact target equality was verified.

A larger ball budget helps only if it permits further splitting. Cora formed 52–53 balls with either budget, so increasing $B_{\max}$ from 64 to 128 left its targets unchanged. CiteSeer formed 63–64 and 63–66 balls, respectively. CoraFull reached both budgets: using 128 balls improved accuracy on all five seeds, with a paired gain of 1.57 ± 0.54 percentage points, and lowered the mean target Brier score from 0.6321 to 0.6026. Finer groups were therefore beneficial on CoraFull in this comparison.

The effects of majority-class weighting and minimum ball size differed across datasets. With $\beta = 1$, CiteSeer improved on all five seeds, by 1.74 ± 0.53 points in paired differences, while the mean changes on Cora and CoraFull were +0.04 and −0.14 points. Reducing $n_{\min}$ to 1 changed accuracy by −1.65, +0.70, and −0.05 points on Cora, CiteSeer, and CoraFull, respectively. Neither change improved all three datasets.

Raising the voting threshold leaves more nodes with their observed labels as targets. At $s_{\min} = 10$, the fractions of training nodes with a positive consensus weight were 56.53%, 58.32%, and 99.13% on Cora, CiteSeer, and CoraFull, respectively. Accuracy changed by −7.27, −5.22, and −0.10 points. The larger losses on Cora and CiteSeer accompanied the lower coverage of consensus supervision. A stricter split threshold, $\delta = 0.10$, prevented splitting on every seed and produced the same single-ball targets as raw X and A^2X. In contrast, $\delta = 0.01$ gave the same targets as the default $\delta = 0.05$ in all 15 runs.

I.2 Error reduction on training targets

Equation (H.5) separates the change in target error into two parts: the reduction on corrupted labels and the increase on initially correct labels. Table I.2 reports both for the default configuration. For each seed, each part is summed over the corresponding nodes and divided by the total number of training nodes, so their difference gives the net reduction in Brier score. The table averages these quantities over five seeds. Clean training labels are used only to evaluate target error and the local class counts examined below.

Table I.2. Decomposition of the reduction in training-target Brier score under uniform 0.4 noise.

Dataset	One-hot label	Soft target	Reduction on corrupted labels	Cost on correct labels	Net reduction
Cora	0.8118	0.4498	0.4034	0.0415	0.3620
CiteSeer	0.8162	0.5352	0.3304	0.0493	0.2811
CoraFull	0.8011	0.6321	0.1960	0.0269	0.1690

In all 15 runs, the reduction on corrupted labels outweighed the increase on correct labels. The targets therefore had lower aggregate error even though some initially correct annotations were perturbed. Random grouping also lowered Brier scores, but its classification accuracy was much lower than that of GB-LCSS (Table I.1). This comparison shows why lower target error alone does not explain the benefit of grouping related nodes.

We also examined the local-class condition in Proposition 1. Using clean labels for this diagnostic, we counted training nodes whose true class had more votes than any other class among the other training nodes in their ball. Averaged over seeds, these fractions were 75.64%, 65.02%, and 36.13% on Cora, CiteSeer, and CoraFull, respectively. The condition is node-specific: a dominant class within a ball can provide useful evidence for members of that class while disagreeing with other members.